Resource Management, Recovery and Recycling

Presentation on Critical Materials

Phosphate Rock

As a critical raw material



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Introduction

- What are critical raw materials?

  Materials that are critical for EU economy

  - The basis of listing as critical raw materials is based on two factors
    - Supply risk(SR)
    - Economic importance(EI)
  - The risks is associated with
    - Low substitution
    - Low recycling rate

Phosphate rock

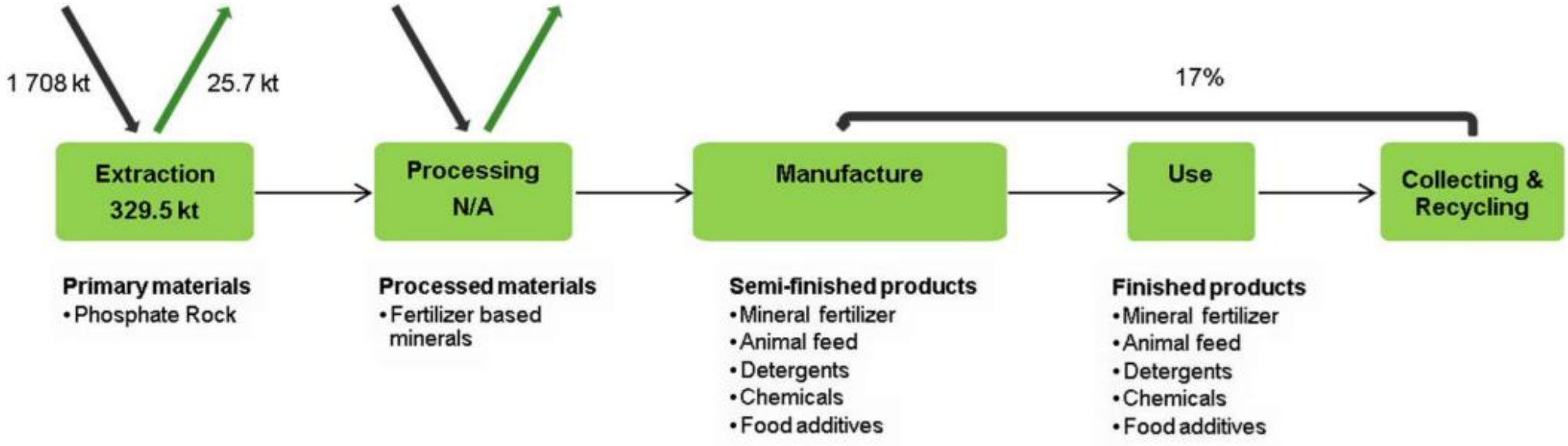
- Mineral deposits

- Extracted and used in agriculture and industry
  - Fertilizer
  - Chemicals or phosphoric acid



Simplified value chain of Phosphate rock

Data in tonnes of P_2O_5 for the EU, averaged over 2012-2016



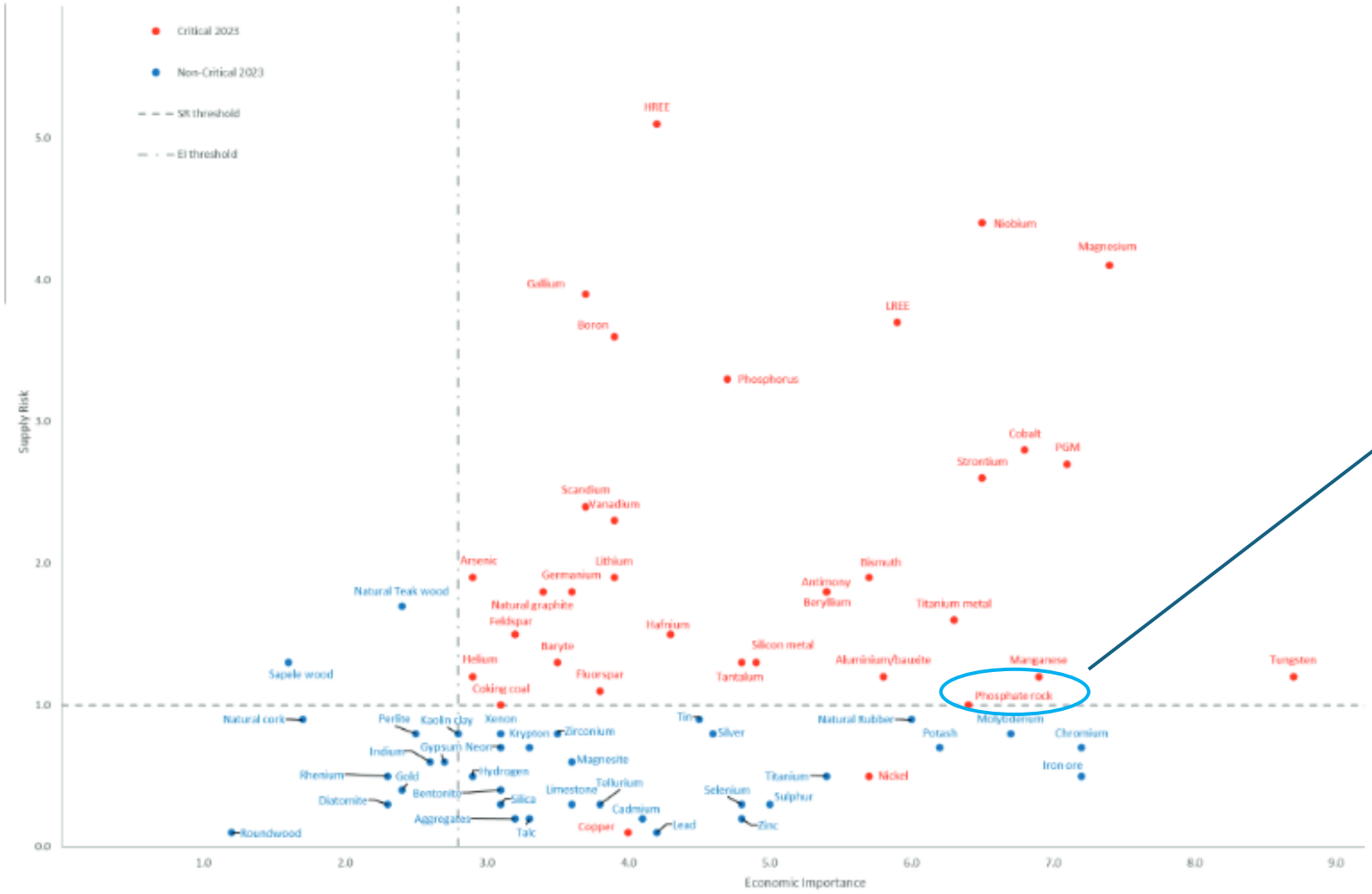
Critical raw materials -2023 list

Screened raw materials in 2023 assessment (new materials in blue)	
Industrial and construction minerals	aggregates, baryte, bentonite, borates, diatomite, feldspar, fluorspar, gypsum, kaolin clay, limestone, magnesite, natural graphite, perlite, phosphate rock, phosphorus, potash, silica sand, sulphur, talc
Iron and ferro-alloy metals	chromium, cobalt, manganese, molybdenum, nickel, niobium, tantalum, titanium, titanium metal, tungsten, vanadium
Precious metals	gold, silver, and Platinum Group Metals (iridium, palladium, platinum, rhodium, ruthenium)
Rare earths	heavy rare earths - HREE (dysprosium, erbium, europium, gadolinium, holmium, lutetium, terbium, thulium, ytterbium, yttrium); light rare earths - LREE (cerium, lanthanum, neodymium, praseodymium and samarium); and scandium
Other non-ferrous metals	aluminium/bauxite, antimony, arsenic, beryllium, bismuth, cadmium, copper, gallium, germanium, gold, hafnium, indium, lead, lithium, magnesium, rhenium, selenium, silicon metal, silver, strontium, tellurium, tin, zinc, zirconium
Bio and other materials	natural cork, natural rubber, natural teak wood, sapele wood, coking coal, hydrogen, helium, roundwood, neon, krypton, xenon

Raw material lists by year

Year	Number of listed CRM	Inclusion of Phosphate rock in the CRM list	Supply risk –SR For PR	Econimic Importance for PR
2011	14	Not assessed	-	-
2014	20	Yes	1.1	5.8
2017	27	Yes	1.0	5.1
2020	30	Yes	1.1	5.6
2023	34	Yes	1.0	6.4

Results of the critiality assessment 2023

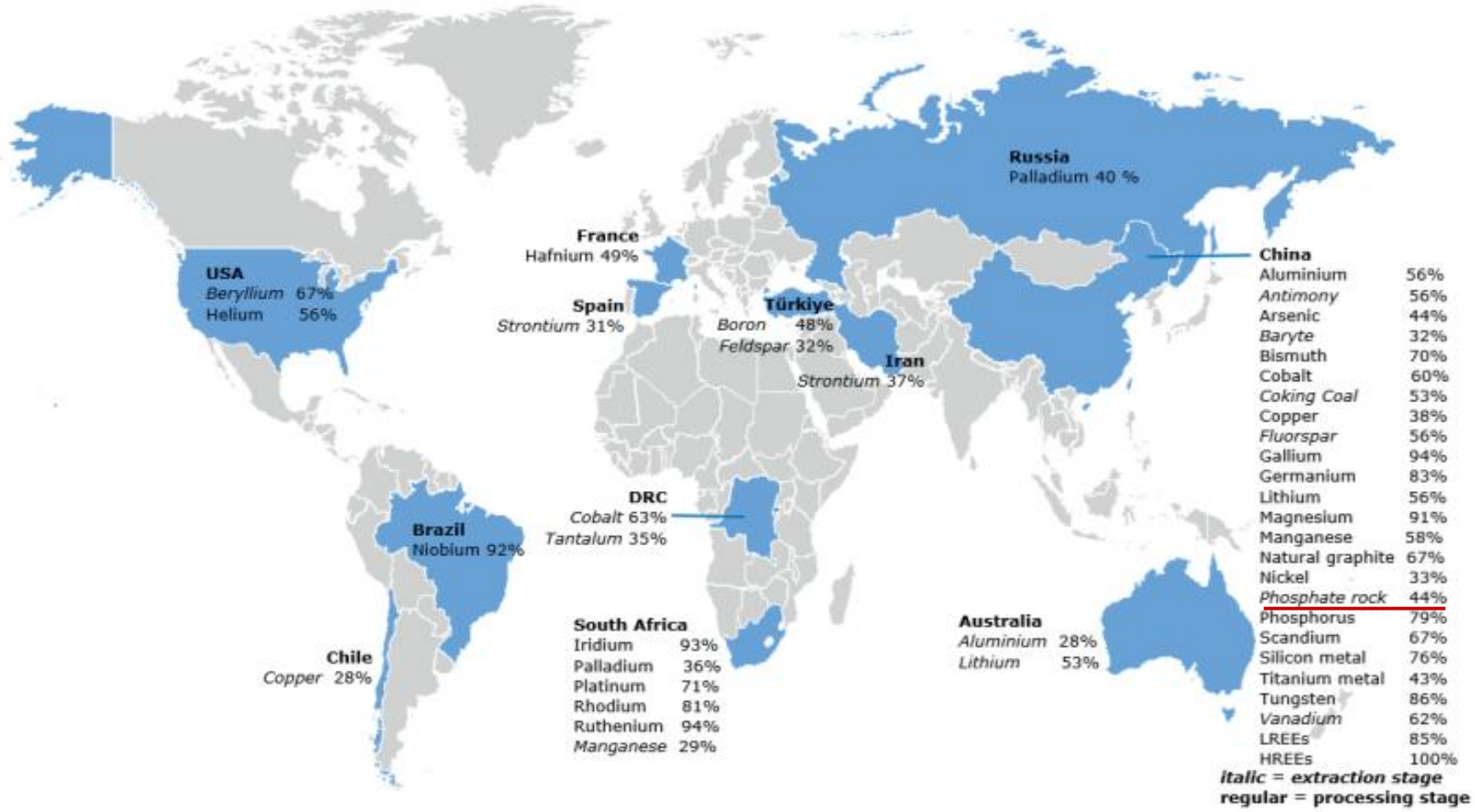


Phosphate rock

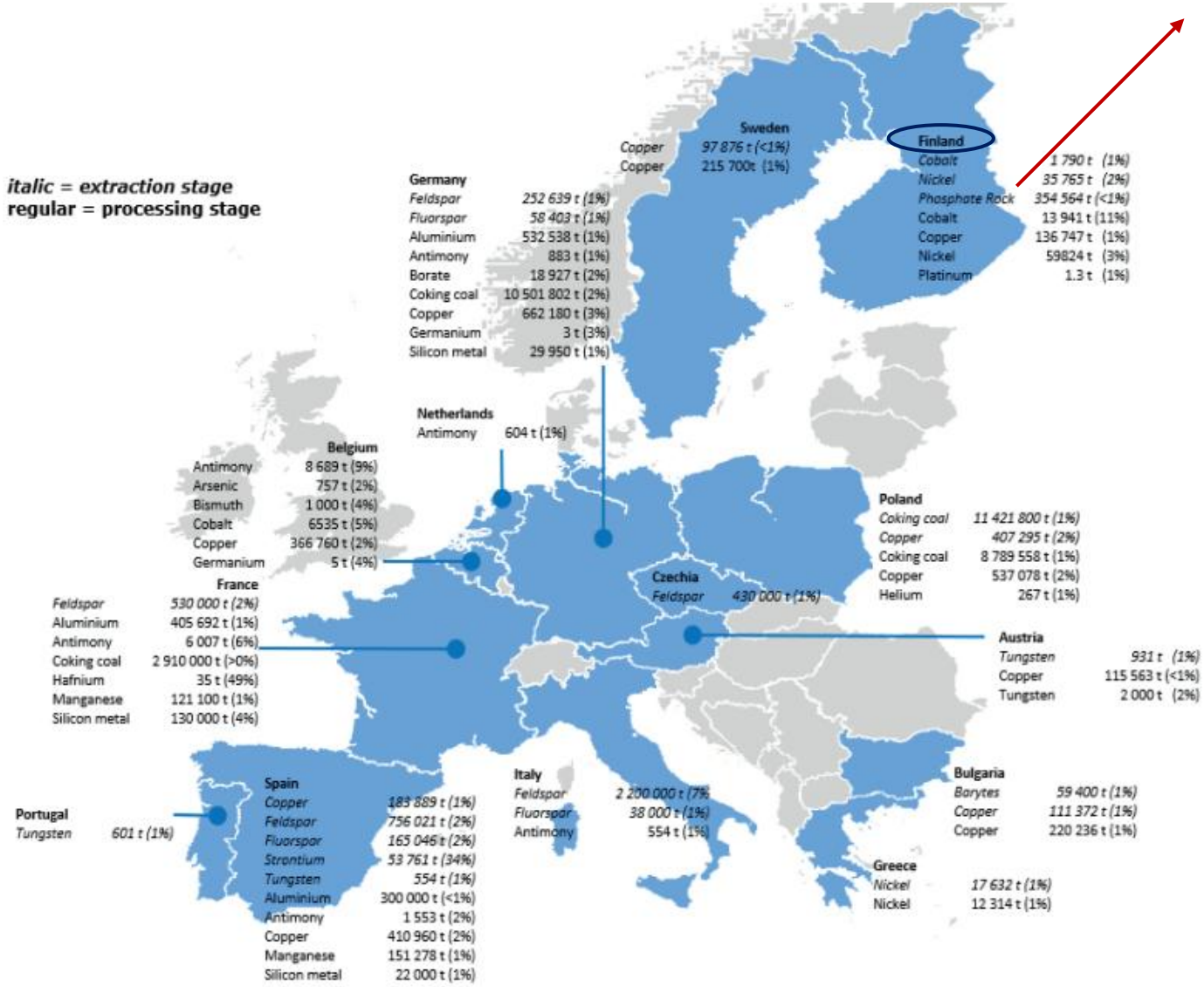
Source: 2023 EU CRM final report

Global Demand and Supply of CRMs

Countries accounting for largest share of global supply of CRM

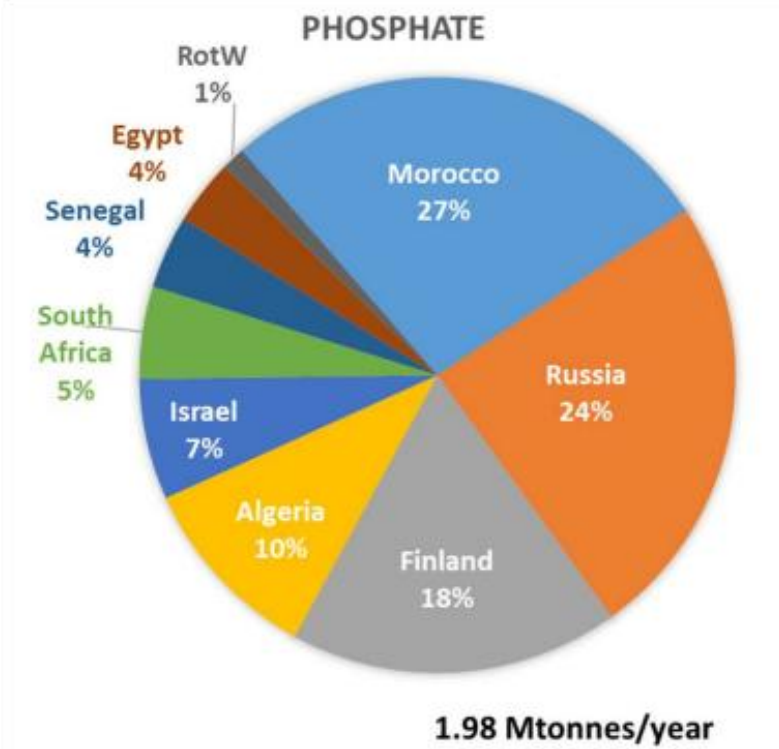


EU-Producers of CRMs (2016-2020)



Major EU sourcing countries of CRMS

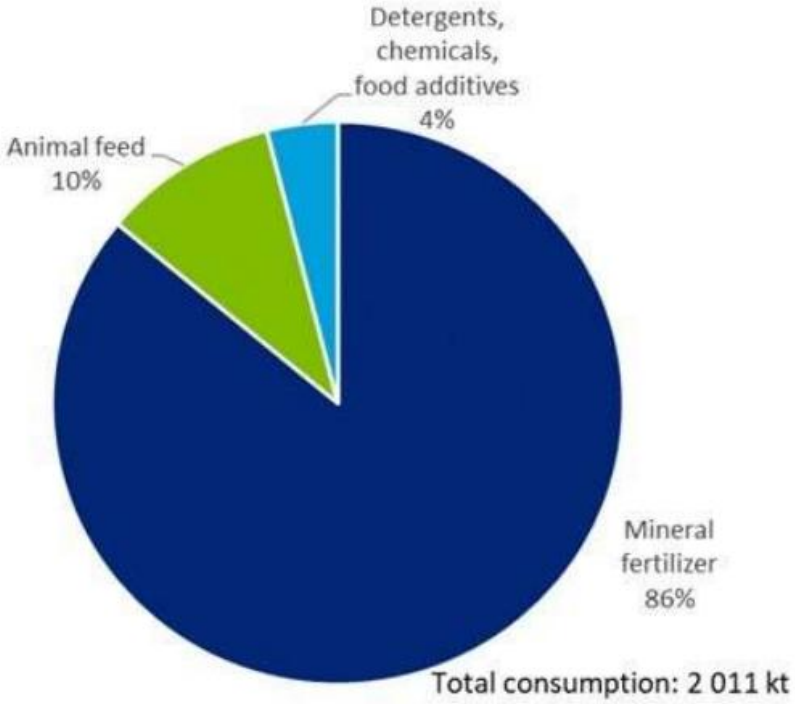
	Material	Stage *	Main EU supplier	Share		Material	Stage *	Main EU supplier	Share
1	aluminium	E	Guinea	63%	27	magnesium	P	China	97%
2	antimony	E	Türkiye	63%	28	manganese	E	S. Africa	41%
3	arsenic	P	Belgium	59%	29	natural graphite	E	China	40%
4	baryte	E	China	45%	30	neodymium	P	China	85%
5	beryllium	E	USA	60%	31	niobium	P	Brazil	92%
6	bismuth	P	China	65%	32	nickel	P	Russia	29%
7	boron	E	Türkiye	99%	33	palladium	P	N/A*	N/A*
8	cerium	P	China	85%	34	phosphate rock	E	Morocco	27%
9	cobalt	E	N/A*	N/A*	35	phosphorus	P	Kazakhstan	65%
10	coking coal	E	Poland	26%	36	platinum	P	N/A*	N/A*
11	copper	E	Poland	19%	37	praseodymium	P	China	85%
12	dysprosium	P	China	100%	38	rhodium	P	N/A*	N/A*
13	erbium	P	China	100%	39	ruthenium	P	N/A*	N/A*
14	europium	P	China	100%	40	samarium	P	China	85%
15	feldspar	E	Türkiye	51%	41	scandium	P	China	67%
16	fluorspar	E	Mexico	33%	42	silicon metal	P	Norway	33%
17	gadolinium	P	China	100%	43	strontium	E	Spain	99%
18	gallium	P	China	71%	44	tantalum	E	Congo, D.R.	35%
19	germanium	P	China	45%	45	terbium	P	China	100%
20	hafnium	P	France	76%	46	thulium	P	China	100%
21	helium	P	Qatar	35%	47	titanium metal	P	Kazakhstan	36%
22	holmium	P	China	100%	48	tungsten	P	China	32%
23	iridium	P	N/A*	N/A*	49	vanadium	E	China	62%
24	lanthanum	P	China	85%	50	ytterbium	P	China	0%
25	lithium	P	Chile	79%	51	yttrium	P	China	100%
26	lutetium	P	China	100%					



EU sourcing of PR 2012-2016

Application areas of PR

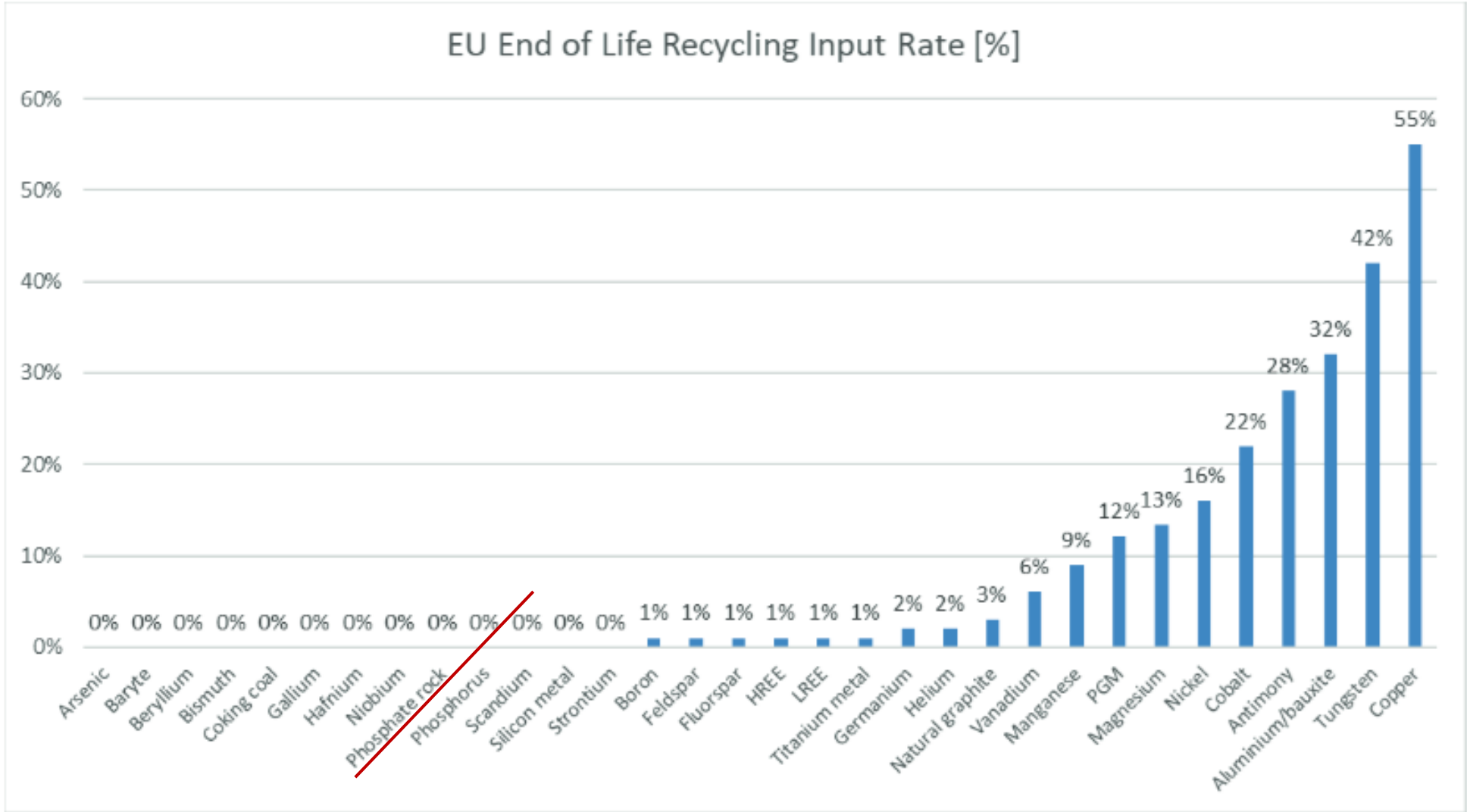
- Production of chemicals (e.g.flame retardants, oil additives, industrial water treatment, emulsifying agents).
- Production of fertilizers
  - The fertiliser industry and related agricultural businesses are considered essential industries in most countries



EU end sourcing of PR (2012-2016)

How to substitute PR and recycling/recovery

- **There are no substitutes for phosphate rock in the production of fertilisers. (ESPP, 2019).**
- In agriculture - phosphate rock can be replaced by secondary sources of phosphorous such as manure, sewage and food waste.
- The future market of phosphate rock is in a great extent controlled by changes in supply and demand of fertilisers used in agriculture and strongly connected with the global **population growth**


EU End of Life Recycling Input Rate [%]
60%
50%
40%
30%
20%
10%
0%
0% 0% 0% 0% 0% 0% 0% 0% 0% 0% 0% 0% 0%
1% 1% 1% 1% 1% 1%
2% 2% 3% 6% 9% 12% 13% 16% 22% 28% 32% 42% 55%
Arsenic
Baryte
Beryllium
Bismuth
Coking coal
Gallium
Hafnium
Niobium
Phosphate rock
Phosphorus
Scandium
Silicon metal
Strontium
Boron
Feldspar
Fluorspar
HREE
LREE
Titanium metal
Germanium
Helium
Natural graphite
Vanadium
Manganese
PGM
Magnesium
Nickel
Cobalt
Antimony
Aluminium/bauxite
Tungsten
Copper

- The International Fertiliser Association forecasts a steady increase in global demand for phosphate until 2025, from 46,3Mt of fertilizer in 2020 up to 52,5Mt in 2025.

- Africa is the fastest growing market for mineral fertilisers, while Latin America is expected to be the region driving the increase in global demand (IFA 2021).

- The consumption of meat is the primary driver of phosphate demand, with an estimated 72% of the global average phosphorus footprint

- There is significant research being made into recovery of phosphorus from wastewater and animal manures.

- These efforts at recovering phosphorus from secondary sources could play a major role in increasing global supply and limit dependence on phosphorus mining

Literature

- Reports - List of Critical Raw Materials for the EU 2014,2017,2020 and 2023
- Study on the review of the list of Critical Raw Materials-2017,2020
- Eu factsheet phosphate rock and Phosphorus 2020